Spring 2026 Midterm 2

The exam is closed book and closed notes. You are allotted one 8.5" x 11", double sided, "cheat sheet" of handwritten notes. You may also use a simple scientific (non-graphing) calculator. Your phone is not allowed as a calculator. Using any resources outside of the aforementioned items is strictly prohibited.

While you take the exam, you are prohibited from discussing the test with anyone. If you are taking the test after your classmates, you are also prohibited from talking to them about the test before you take it. Evidence of cheating may result in a 0 on the exam and be reported to the Student Conduct Board.

First and Last Name (print clearly): _____

Student ID Number (print clearly): _____

Please read the following statement and sign below to indicate that you understand the policies for this exam

You must show your student ID when you submit your test.

UC Berkeley Honor Code
"As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others." Please carefully read the statements below, and indicate your understanding and intent to adhere to the UC Berkeley Honor code by signing your name in the space below. I agree not to engage in any of the following behaviors:

- Copying or attempting to copy from others during an exam or on an assignment.
- Communicating answers with another person during an exam.
- Pre-programming a calculator or other personal electronic device to contain answers, or using other unauthorized information for exams.
- Using unauthorized materials, i.e. prepared answers.
- Allowing others, or utilizing software like generative AI, to do an assignment or a portion of an assignment for you, including the use of a commercial term-paper service.
- Submitting the same assignment for more than one course without prior approval of all the instructors involved.
- Collaborating on an exam or assignment with any other person without prior approval from an instructor.
- Taking an exam for another person or having someone take an exam for you.
- Altering a previously graded exam or assignment for the purpose of a grade appeal or of gaining points in a re-grading process.
- Submitting an electronic file the student knows to be unreadable or corrupted instead of a completed assignment.

Signature:

Sign your name: _____

INSTRUCTIONS:

Handwrite your responses using a pencil or pen in the space provided. Give your responses ONLY in the space provided. Do not write your responses outside of the provided spaces. The additional space provided, including space on the back side of the exam can be used as scratch paper but will not be graded.

Phones should be turned OFF prior to the start of the exam and secured in your backpack or another secure location. Do not leave your phone or other electronic devices out. If you need to leave the room for any reason during the exam please flag a GSI and let them know prior to exiting the room. Upon leaving the exam room, you must leave your electronic device(s) with the GSIs. Time will still accrue when you leave the room.

The length of the midterm is 50 minutes. If you finish early and are satisfied with your work you may leave early. Hand in your exam to a GSI, who will verify that they received it, with your student ID.

- Unless otherwise specified in the question, format your answers according to the following guidelines:
  - present your answers rounded to two decimal places
  - present proportions as decimal values rounded to 2 decimal places rather than percents (ex: 0.41 rather than 41.30%)
- All logarithms ("logs") are the natural logarithm (base e; $log_e(x)$; $ln(x)$)

Exam Format:

Short Format Questions: Q1-6 [15 points]
Long Format Question: Q7 [7 points]
Long Format Question: Q8 [11 points]
Long Format Question: Q9 [12 points]
Extra credit: Q10 [2 points]
Optional Feedback Question

Short Format Questions: Please select only one answer choice.

Q1 [2 points] True or False: If two events A and B are independent, then knowing that B occurred does not change the probability that A occurred. Therefore, $P(A|B) = P(B|A)$

- ☐ A. True
- ☐ B. False

Q2 [2 points] A campus health survey reports the following among students: 30% report exercising regularly, 45% report eating breakfast daily, and 18% report both exercising regularly and eating breakfast daily. What is the probability that a randomly selected student reports at least one of these behaviors (exercising regularly or eating breakfast daily)?

- ☐ A. 0.75
- ☐ B. 0.57
- ☐ C. 0.93
- ☐ D. 0.18

Q3 [2 points] Public health officials are monitoring avian influenza (H5N1) after several human infections were detected among agricultural workers in the United States. A screening test for H5N1 has:

- Sensitivity = 95%
- Specificity = 95%

Suppose the infection affects 0.5% of the population being tested.

A health official states: "Because the test has high sensitivity and specificity, most people who test positive will actually have the infection." Which of the following statements is most accurate?

- ☐ A. The statement is correct because high sensitivity ensures most positive results are true positives
- ☐ B. The statement is correct because both sensitivity and specificity are high
- ☐ C. The statement may be incorrect because when a disease is rare, false positives can outnumber true positives
- ☐ D. The statement is incorrect only if the sensitivity is lower than the specificity

Q4 [2 points] A city health department tracks the number of reported West Nile virus cases in a metropolitan area. Over the past five summers, the area has averaged 2.3 cases per week during peak mosquito season. An epidemiologist wants to calculate the probability that exactly 5 cases are reported in a given week. Which probability distribution is most appropriate?

- ☐ A. Normal distribution, because the number of cases per week will be approximately symmetric and centered around the average of 2.3 cases per week
- ☐ B. Binomial distribution, because each resident either does or does not contract West Nile virus
- ☐ C. Binomial distribution, because each resident has a fixed probability of infection each week
- ☐ D. Poisson distribution, because West Nile cases are rare events occurring at a constant average rate, over a fixed interval of time, with no fixed upper bound on the count

Q5 [2 points] Researchers compare newborn birth weights across two different populations.

- Baby A weighs 3.28 kg in a population where the mean birth weight is 3.10 kg and the standard deviation is 0.18 kg.
- Baby B weighs 4.02 kg in a population where the mean birth weight is 3.75 kg and the standard deviation is 0.24 kg.

Which baby's birth weight is more unusual compared to babies in their own population?

☐ A. Baby A
☐ B. Baby B
☐ C. They are equally unusual
☐ D. Cannot be determined from the information given

Q6 [5 points]

Creatine is a popular dietary supplement used by athletes to improve strength. Researchers measure the one-rep max bench press weight (kg) among recreational athletes who regularly take creatine.

In the population of recreational athletes who use creatine, bench press weights are right-skewed with a mean of 95 kg and a standard deviation of 20 kg.

A researcher takes a random sample of 36 athletes and records their bench press weights.

Q6.1 [1.5 points] What are the mean and standard error of the sampling distribution of the sample mean for n = 36?

☐ A. The sampling distribution has mean 95 and standard deviation 20
☐ B. The sampling distribution has mean 95 and standard error $\frac{20}{\sqrt{36}}$
☐ C. The sampling distribution has mean $\frac{95}{36}$ and standard deviation $\frac{20}{\sqrt{36}}$
☐ D. The sampling distribution has mean 95 and standard deviation $\frac{20}{36}$

Q6.2 [1.5 points] Researchers repeat this sampling process many times. For each sample of 36 athletes, they calculate the sample mean bench press weight. Approximately what proportion of these sample means will fall between 91.7 kg and 98.3 kg?

☐ A. 0.975
☐ B. 0.95
☐ C. 0.68
☐ D. 0.34

Q6.3 [2 points] Which of the following statements about the sampling distribution of the sample mean is TRUE?

☐ A. Because the population distribution is right-skewed, the sampling distribution of the mean is also right-skewed
☐ B. The sampling distribution of the mean will be approximately normal because the sample size is large enough for the Central Limit Theorem to apply
☐ C. The sampling distribution of the mean has a standard deviation equal to the population standard deviation (20 kg)
☐ D. Increasing the sample size would increase the standard error of the sample mean

Long Format Questions

Q7 [7 points]

Abstract Environmental exposures such as phthalates–commonly found in plastics and personal care products–have been associated with potential neurocognitive effects. Researchers aim to evaluate whether higher phthalate exposure is associated with slower cognitive processing among young adults.

Methods Researchers recruit adults aged 18–35 and measure urinary phthalate metabolite levels. Participants are categorized into two groups:
- High exposure group (n = 120)
- Low exposure group (n = 120)

Cognitive processing speed is assessed using a computerized reaction time test (measured in milliseconds).

Results:
- High exposure group: mean = 285 ms, standard deviation = 35 ms
- Low exposure group: mean = 260 ms, standard deviation = 30 ms
- A higher reaction time indicates slower cognitive processing

Q7.1 [2 points] State an appropriate null and alternative hypothesis in the context of this question using statistical notation (e.g., $\mu = \dots$)

H_0 :

H_A :

Q7.2 [2 points] Calculate the 95% confidence interval for the mean reaction time in the high exposure group. Do not round intermediate values. Show your work and round to two decimal places.

Final Answer (round to 2 decimals):

Q7.3 [1 point] Interpret the confidence interval from 7.2 in the context of this question (1 sentence).

Q7.4 [1 point] True or False: This test has a two-tailed alternative hypothesis.

☐ A. True
☐ B. False

Q7.5 [1 point] The resulting p-value for this comparison is 0.003. Based on these results, you decide to _____ the null hypothesis.

☐ A. Reject
☐ B. Fail to reject

Q8 [11 points]

Read the following abstract and answer the questions below.

Sunburn is a major risk factor for skin cancer, and sunscreen use is widely recommended as a protective behavior. Researchers conducted a cross-sectional survey to study sun protection behaviors among adults in the United States. A random sample of adults was asked about sunscreen use and sunburn experiences during the previous summer. In the survey, 65% of participants reported regularly using sunscreen with SPF 30 or higher when outdoors, 27% reported experiencing at least one sunburn during the previous summer, and 12% reported both regularly using sunscreen and experiencing at least one sunburn.

Let

- S = individual regularly uses sunscreen
- B = individual experienced at least one sunburn during the previous summer

[Hint] You may want to draw a Venn Diagram or other visualization.

Q8.1 [3 points] What is the probability that a randomly selected individual experienced a sunburn but did not regularly use sunscreen? Show your work.

Final Answer (round to 2 decimals):

Q8.2 [4 points] What is the probability that a randomly selected individual neither regularly used sunscreen nor experienced a sunburn last summer? Show your work.

Final Answer (round to 2 decimals):

Q8.3 [4 points] Are sunscreen use and experiencing a sunburn independent events in this population? Show your work and justify your answer using probabilities.

Q9 [12 points]

Commercial IV electrolyte therapy services claim that their treatments improve recovery after strenuous exercise. Researchers evaluate this claim in a randomized study of 180 marathon runners immediately after a race. Runners are randomly assigned to receive either: an IV electrolyte treatment, or a placebo saline IV.

Researchers would like to test whether the IV electrolyte treatment improves recovery compared with the historical recovery rate. They define successful recovery as reporting minimal muscle soreness 48 hours after the race. Previous studies suggest that approximately 28% of runners recover within 48 hours without treatment. In the study, 54 runners in the treatment group (n = 90) reported successful recovery.

Q9.1 [2 points] Define the null hypothesis and alternative hypothesis for this study.

H_0 :

H_A :

Q9.2 [3 points] Assume the true probability of recovery is 28%. If 90 runners are treated, answer the following:

a) What is the expected number of recoveries? Show your work.

Final Answer (round to 2 decimals):

b) What is the standard deviation of the number of recoveries?

Final Answer (round to 2 decimals):

Q9.3 [3 points] Using a normal approximation to the binomial, calculate the z-score for the observed result of 54 recoveries in the treatment group. Show your work.

Final Answer (round to 2 decimals):

Q9.4 [2 points] Suppose the researchers want to increase the statistical power of the test. Describe ONE change to the study design they could use to increase the statistical power of the test (1 sentence).

Q9.5 [2 points] Write the line of R code that calculates the probability of observing a z-score greater than or equal to the one you calculated in Q9.3.

Q10 [2 points] Extra Credit: The amount of time visitors spend hiking in the Tilden Park Botanical Garden is approximately normally distributed with a mean of 60 minutes and a standard deviation of 15 minutes. A park ranger randomly observes 10 visitors entering the garden. What is the probability that exactly 1 visitor spends more than 80 minutes hiking in the garden? Show your work.

Final Answer (round to 2 decimals):

Optional feedback

Take a moment to let us know any issues that came up on the exam.

This space is intended for scratch paper, we will NOT be grading this page.